CAO – Question Bank – Solutions

2 Marks

1. **Define number system.**
   A number system is a writing system for expressing numbers; it defines a set of symbols and rules to represent numbers in a consistent way. Common examples include binary, decimal, octal, and hexadecimal systems.

2. **Explain programmed I/O.**
   Programmed I/O is a method where the CPU has direct control over I/O operations, polling devices to check if they are ready. The CPU is actively involved in transferring data, which can lead to inefficiencies as it remains idle during data transfer.

3. **What is direct mapping?**
   Direct mapping is a cache memory technique where each block of main memory maps to only one possible cache line. This simplifies cache design but can lead to more frequent cache misses if multiple frequently accessed data blocks map to the same cache line.

4. **List out types of computer organization.**
   The types include:
   - Single Accumulator Organization
   - General Register Organization
   - Stack Organization

5. **What is cache memory?**
   Cache memory is a high-speed memory located between the CPU and main memory. It temporarily stores frequently accessed data and instructions, reducing access time and improving processing speed.

6. **Types of interleaved memory.**
   Types of interleaved memory include:
   - Low-order interleaving
   - High-order interleaving

7. **Concept of pipelining.**
   Pipelining is a technique where multiple instruction phases (fetch, decode, execute, etc.) are overlapped in a pipeline, allowing simultaneous execution of multiple instructions. This increases CPU throughput and performance.

8. **Write DMA.**
   DMA stands for Direct Memory Access, which allows hardware devices to access main memory independently of the CPU, enhancing data transfer efficiency.

9. **Give examples of hardware and software.**
   - Hardware: CPU, RAM, Hard Disk
   - Software: Operating System, Word Processor, Web Browser

10. **Define interleaved memory.**
    Interleaved memory is a technique where memory is divided into modules and accessed in an alternating pattern to improve speed by overlapping memory access times.

11. **Advantages of direct mapping.**
    - Simple and easy to implement

- Lower cost due to straightforward mapping
- Quick access time for the mapped cache line

12. **Draw microinstruction code format.**
    Microinstruction formats vary, but a basic format includes fields for:
    - Control signals
    - Opcode or instruction code
    - Condition field (for branching)
13. **Define RISC.**
    RISC (Reduced Instruction Set Computer) is a CPU design that uses a small, highly optimized set of instructions that execute in a single clock cycle to achieve higher performance.
14. **State the functional units of a basic computer system.**
    Functional units include:
    - Input Unit
    - Output Unit
    - Memory Unit
    - Arithmetic and Logic Unit (ALU)
    - Control Unit
15. **Explain Set-Associative Mapping.**
    Set-associative mapping is a cache memory method where each memory block can be mapped to any of a specific set of cache lines, combining benefits of direct and fully associative mapping.
16. **Write interleaved memory.**
    Interleaved memory divides memory into modules that the CPU can access simultaneously to increase performance by allowing parallel access.
17. **Write characteristics of Read-Only Memory (ROM).**
    - Non-volatile: retains data without power
    - Pre-programmed with data or instructions
    - Used for firmware in computers and electronic devices
18. **Give example of direct addressing mode.**
    In direct addressing mode, the operand's memory address is specified directly in the instruction. Example: LOAD A, where A is the memory address of the operand.
19. **Define virtual memory.**
    Virtual memory is a memory management technique that uses a section of secondary storage as if it were additional RAM, enabling larger programs to run on systems with limited physical memory.
20. **Define interrupt.**
    An interrupt is a signal sent to the CPU, indicating an event that requires immediate attention, temporarily halting the current process to execute a specific service routine.
21. **State the advantages of computer system.**
    - High speed and accuracy in processing
    - Reliable data storage and retrieval
    - Automation of repetitive tasks

- Connectivity and networking capabilities

22. **Identify the following diagram and write the correct name of addressing mode.**
The exact diagram is not provided, but common addressing modes include direct, indirect, immediate, and indexed addressing.

23. **List out types of addressing modes.**
    - Immediate addressing
    - Direct addressing
    - Indirect addressing
    - Register addressing
    - Indexed addressing

24. **Write application of ROM.**
    - Storing firmware or system software
    - Booting programs in computers
    - Embedded applications in appliances

25. **List types of buses.**
    - Data bus
    - Address bus
    - Control bus

26. **Define flash memory.**
Flash memory is a type of non-volatile storage that can be electronically erased and reprogrammed, commonly used in USB drives, SSDs, and memory cards.

27. **Give example of register addressing mode.**
In register addressing mode, operands are in CPU registers. Example: MOV AX, BX, where AX and BX are registers.

28. **Define decimal number system.**
The decimal number system is a base-10 system that uses ten digits (0 to 9) to represent numbers, commonly used in everyday arithmetic.

29. **Draw the flow of instruction cycle.**
The instruction cycle typically includes:
    - Fetch: Retrieving the instruction from memory
    - Decode: Interpreting the instruction
    - Execute: Performing the operation

4 Marks

1. **Compare virtual memory and cache memory.**
    - *Virtual Memory*: Extends physical memory by using disk space, allowing larger programs to run on limited physical memory. It uses paging or segmentation to manage memory space.
    - *Cache Memory*: A small, high-speed memory between the CPU and main memory, storing frequently accessed data to reduce access time and improve performance.
    - *Key Differences*: Cache memory improves CPU speed, while virtual

memory enhances memory capacity. Cache is fast and closer to the CPU, whereas virtual memory relies on slower secondary storage.

2. **Explain different addressing modes in Computer Architecture.**
    - *Immediate Addressing*: Operand is specified within the instruction.
    - *Direct Addressing*: Instruction holds the memory address of the operand.
    - *Indirect Addressing*: Address field points to a memory location, which holds the address of the operand.
    - *Register Addressing*: Operand is in a CPU register.
    - *Indexed Addressing*: Uses an index register to modify the address in the instruction.
3. **Explain hardwired and microprogrammed control unit.**
    - *Hardwired Control Unit*: Implements control logic with fixed electronic circuits, leading to faster operation but less flexibility.
    - *Microprogrammed Control Unit*: Uses a sequence of microinstructions stored in memory to control operations, providing flexibility but slower execution compared to hardwired units.
4. **Examine X=(A+B)∗(C−D)/EX = (A + B) * (C - D) / EX=(A+B)∗(C−D)/E using 3, 2, and 0 Address Instruction format.**
    - *3 Address Format*: ADD A, B, T1; SUB C, D, T2; MUL T1, T2, T3; DIV T3, E, X
    - *2 Address Format*: LOAD A; ADD B; STORE T1; LOAD C; SUB D; STORE T2; LOAD T1; MUL T2; DIV E; STORE X
    - *0 Address (Stack) Format*: PUSH A; PUSH B; ADD; PUSH C; PUSH D; SUB; MUL; PUSH E; DIV; POP X
5. **Write a short note on Booth's Algorithm.**
   Booth's Algorithm is a multiplication algorithm for signed binary numbers. It reduces the number of required additions when consecutive 1s or 0s are encountered by encoding the binary multiplier. The algorithm is efficient for signed two's complement numbers and is commonly used in computer processors.
6. **Draw single and double precision format of IEEE floating-point representation.**
    - *Single Precision*: 1-bit sign, 8-bit exponent, and 23-bit mantissa.
    - *Double Precision*: 1-bit sign, 11-bit exponent, and 52-bit mantissa. These formats follow IEEE 754 standards for representing

floating-point numbers in binary.

7. **Compare RAM and ROM.**
    - *RAM (Random Access Memory)*: Volatile memory, data is lost when power is off, used for temporary data storage and quick access during runtime.
    - *ROM (Read-Only Memory)*: Non-volatile memory, retains data without power, typically used to store firmware and system initialization instructions.
    - *Key Difference*: RAM is writable during normal operations, while ROM is preprogrammed and cannot be modified easily.
8. **Explain Format of Microinstruction in Computer Architecture.**
   A microinstruction includes fields like:
    - *Control Signals*: Define specific operations in the CPU.
    - *Condition Field*: Specifies conditions for branching.
    - *Opcode Field*: Indicates operation code for execution. This format allows sequential or conditional control over microoperations.
9. **Express the term Addressing modes, Interrupts, Processor Bus, and Interfacing.**
    - *Addressing Modes*: Techniques to specify operand locations.
    - *Interrupts*: Signals that pause CPU operations to handle events.
    - *Processor Bus*: Communication pathways for data transfer.
    - *Interfacing*: Linking different hardware units for communication.
10. **Examine (11011)2−(10011)2(11011)_2 - (10011)_2(11011)2−(10011)2 using 1's complement and 2's complement method.**
    - *1's Complement*: Convert 10011 to 1's complement: 01100 Add 11011 + 01100 = 100111 (discard extra 1), result = 00111 or 7107_{10}710
    - *2's Complement*: Convert 10011 to 2's complement: 01101 Add 11011 + 01101 = 101000 (discard extra 1), result = 01000 or 8108_{10}810
11. **Draw Booth's Algorithm flowchart.**
    The flowchart typically includes steps for:
    - Initialize registers
    - Check LSB and perform actions based on pairs (0-1, 1-0)
    - Shift and adjust accumulator

- Repeat until count is zero

12. **Difference between primary and secondary memory.**
    - *Primary Memory*: Fast, volatile memory (e.g., RAM) used by the CPU to store data temporarily.
    - *Secondary Memory*: Slower, non-volatile memory (e.g., HDD, SSD) for long-term data storage.
    - Primary memory is more expensive per bit compared to secondary memory.
13. **Explain static and dynamic RAM.**
    - *Static RAM (SRAM)*: Stores data as long as power is supplied without needing to refresh, faster but more expensive, used in cache memory.
    - *Dynamic RAM (DRAM)*: Needs regular refreshing to maintain data, slower but cheaper, used in main memory.
14. **Discuss the following terms: Memory delays, Branch delays, Processor bus, and Interfacing.**
    - *Memory Delays*: Latency in accessing data from memory.
    - *Branch Delays*: Delay in instruction execution due to branching.
    - *Processor Bus*: Communication link for data transfer within the CPU.
    - *Interfacing*: Connecting different hardware components for effective communication.
15. **Examine addition of (+15) and (-6) using 2's complement method.**
    - *+15 in binary*: 01111
    - *-6 in 2's complement*: 11010
    - Add: 01111 + 11010 = 01001, which represents +910+9_{10}+910.
16. **Write a short note on format of Microinstruction in Computer Architecture.**
    Microinstruction format consists of fields to specify:
    - *Opcode*: Indicates the type of operation.
    - *Control Signals*: Details for CPU control actions.
    - *Next Address*: Defines next microinstruction location for conditional or sequential execution.
17. **Draw a typical CPU organization.**
    Typical CPU organization includes:
    - ALU (Arithmetic Logic Unit)
    - Control Unit
    - Registers
    - Buses (Data, Address, Control)

7 Marks

Here are solutions for the 7-mark questions in your question bank:

1. **Explain memory management requirements.**
   Memory management is essential for efficiently utilizing memory resources, and it involves:
   - *Relocation*: Adjusting program memory references to match physical memory addresses.
   - *Protection*: Ensuring processes do not interfere with each other's memory.
   - *Sharing*: Allowing multiple processes to access the same memory region when needed.
   - *Logical Organization*: Managing memory in a logical manner, like segmentation.
   - *Physical Organization*: Efficiently organizing physical memory. These features help ensure stability, security, and efficiency in managing system memory.
2. **Explain non-restoring algorithm with example.**
   The non-restoring algorithm is a division algorithm used for signed binary numbers, where subtraction and addition are alternated based on the sign of the remainder:
   - Example: For dividing 101121011_210112 by 001120011_200112,
     - Initialize the dividend and divisor.
     - Perform successive subtraction or addition based on the sign.
     - Adjust results to derive quotient and remainder. This algorithm is used to handle division without repeatedly restoring the remainder.
3. **What is RISC & CISC with characteristics?**
   - *RISC (Reduced Instruction Set Computer)*: Focuses on a limited set of simple instructions that execute quickly. Characteristics include:
     - Single-cycle execution for most instructions
     - Large number of general-purpose registers
     - Simple addressing modes
   - *CISC (Complex Instruction Set Computer)*: Has a wide variety of complex instructions, some taking multiple cycles. Characteristics include:
     - Fewer registers
     - Complex addressing modes
     - Multi-cycle instructions
4. **Multiplication of two signed numbers (10111)2(10111)_2(10111)2 and (10011)2(10011)_2(10011)2 by a**

suitable algorithm.
To multiply signed binary numbers, we can use Booth's algorithm:

- Convert the binary numbers to two's complement if necessary.
- Apply Booth's encoding to minimize the number of additions.
- Add and shift based on Booth's rules until completion. The result will be in two's complement form, indicating the final product.

5. **Discuss parallel processor system in computer architecture.**
Parallel processor systems utilize multiple processors to execute tasks simultaneously, increasing computational speed and efficiency. Types include:

- *SISD (Single Instruction Single Data)*: Traditional single-core processing.
- *SIMD (Single Instruction Multiple Data)*: Executes the same instruction across multiple data points (used in vector processing).
- *MIMD (Multiple Instruction Multiple Data)*: Different processors perform different instructions on different data.
- *MISD (Multiple Instruction Single Data)*: Rare, used in specialized systems.
Parallel processing benefits include faster processing, improved fault tolerance, and the ability to handle large datasets effectively.

6. **Explain the concept of pipelining. How it improves the execution of program over sequential execution?**
Pipelining divides the execution of instructions into separate stages (e.g., fetch, decode, execute) that can be processed concurrently. This allows overlapping of tasks, increasing instruction throughput:

- *Improvement*: In sequential execution, each instruction is processed individually. In pipelining, multiple instructions progress simultaneously in different pipeline stages, reducing total execution time.
- *Example*: In a 5-stage pipeline, 5 instructions can be processed at once, with one instruction completing every clock cycle after the pipeline is full.

7. **What is programmed I/O, I/O modules?**

- *Programmed I/O*: A method where the CPU manages I/O data transfer by actively polling the I/O device, which can slow down CPU efficiency.
- *I/O Modules*: These are interfaces between the CPU and peripheral devices, handling data transfer, control, and communication, which can be improved through Direct Memory Access (DMA) or interrupt-driven I/O to

minimize CPU involvement.

8. **Explain fixed-point representation with a suitable example.**
   Fixed-point representation is a way of storing numbers where a fixed number of digits represent the integer and fractional parts. It is commonly used when decimal precision is required but floating-point is unnecessary:
   - *Example*: For an 8-bit number with 4 integer and 4 fractional bits, 0010.11000010.11000010.1100 represents 2.752.752.75 in decimal.
   - *Limitation*: Fixed-point is less flexible for a wide range of values but suitable for applications requiring consistent precision, such as digital signal processing.
9. **Examine (1269.14)10(1269.14)_{10}(1269.14)10 in a Single Precision and Double Precision format.**
   - *Single Precision*: 1-bit sign, 8-bit exponent, 23-bit mantissa.
   - *Double Precision*: 1-bit sign, 11-bit exponent, 52-bit mantissa. For both, convert 1269.141269.141269.14 to binary, normalize it, and encode the result based on IEEE 754 floating-point format.
10. **What is Single Bus Architecture?**
    In single bus architecture, a single bus serves as a common communication pathway among CPU, memory, and I/O devices. It is simpler and cost-effective but can lead to bottlenecks as only one data transfer can occur at a time.
11. **Analyze (703.63)10(703.63)_{10}(703.63)10 with Single Precision and Double Precision format.**
    - Convert 703.63703.63703.63 to binary.
    - Normalize and adjust exponent and mantissa fields according to IEEE 754 single and double precision specifications.
12. **Examine (1A5DB)16(1A5DB)_{16}(1A5DB)16 to binary, octal, and decimal number systems.**
    - *Binary*: Convert each hex digit to its 4-bit binary equivalent.
    - *Octal*: Group the binary representation into sets of three bits.
    - *Decimal*: Convert the hexadecimal number directly using positional values.
13. **Discuss the functioning of microprogrammed control unit.**
    The microprogrammed control unit manages CPU operations through a sequence of microinstructions stored in a control memory. Each instruction is divided into microinstructions specifying control signals:
    - *Fetch*: Loads the next microinstruction.

- *Decode*: Interprets and signals the control lines.
- *Execute*: Manages data flow based on decoded microinstructions. Microprogramming offers flexibility, allowing complex instructions to be implemented as sequences of simpler operations.

12 Marks

1. **Rewrite about the following:**

a) Programmed I/O:
Programmed I/O is a method where the CPU is responsible for managing all input and output operations. The CPU actively checks the status of the I/O device by repeatedly polling it to see if it is ready for data transfer. While simple to implement, this approach is inefficient for high-speed devices since it keeps the CPU busy in a loop, waiting for the I/O device, which can slow down overall system performance.

b) Vector Processing:
Vector processing involves executing a single operation on multiple data points at once, particularly beneficial in scientific computations that require large data handling. Vector processors are equipped with instructions to perform mathematical operations on entire arrays (vectors) of data in one go, enhancing performance for tasks like matrix calculations, 3D graphics, and scientific simulations.

c) I/O Modules:
I/O modules serve as intermediaries between the CPU and peripheral devices, managing the data flow, control signals, and timing required for effective communication. They often include methods like interrupt-driven I/O, which allows the CPU to continue other tasks until the I/O device is ready, and DMA (Direct Memory Access), where the module handles data transfers directly, freeing the CPU.

d) Control Unit:
The control unit in the CPU interprets instructions and generates control signals to coordinate CPU operations. It controls data flow within the CPU and between the CPU, memory, and I/O devices, ensuring instructions are executed in sequence. Control units can be hardwired (fast but inflexible) or microprogrammed (flexible and easier to modify).

e) Types of Parallel Processor Systems:

SISD (Single Instruction Single Data): Single-core processor executing one instruction on one data element at a time.

SIMD (Single Instruction Multiple Data): Same instruction applied simultaneously across multiple data points, useful in vector processing.

MIMD (Multiple Instruction Multiple Data): Multiple processors execute different instructions on

different data points, common in multiprocessor systems.

MISD (Multiple Instruction Single Data): Multiple instructions operate on the same data, though rarely used in practical systems.

f) Parallel Processing:
Parallel processing uses multiple processors to handle separate tasks simultaneously, increasing computational speed. It is especially useful for applications like simulations, data processing, and large-scale computations. By dividing work across processors, tasks are completed faster, enhancing system throughput.

g) Pipelining Issues:
Pipelining splits instruction execution into stages to improve throughput. However, it faces issues:

Data hazards: Arise when instructions depend on previous ones.

Control hazards: Occur with branching instructions that disrupt pipeline flow.

Structural hazards: When hardware resources are insufficient to support overlapping instructions. Handling these issues requires techniques like forwarding, branch prediction, and pipeline stalling.

h) Microinstruction Sequencing Techniques:
Microinstruction sequencing arranges control signals for efficient CPU operation. Techniques include:

Horizontal Sequencing: Each control signal is individually coded, allowing concurrent operations but requiring more bits.

Vertical Sequencing: Uses a more compact form, where each microinstruction includes fewer signals per cycle. Effective sequencing controls the flow and branching of microinstructions based on conditions and next-instruction pointers.

i) Displacement Addressing Mode:
In displacement addressing, an effective address is calculated by adding a base address in a register with an offset or displacement specified in the instruction. It's useful in accessing elements in arrays, tables, or data structures.

j) Superscalar Processor:
Superscalar processors execute multiple instructions per clock cycle using multiple pipelines. These processors fetch, decode, and execute several instructions simultaneously, improving overall processing speed by utilizing instruction-level parallelism.

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2. **Evaluate the Division of 17 and 3 Using the Restoring Algorithm**

The restoring division algorithm performs binary division by iteratively subtracting or restoring based on the remainder's sign.

Convert 171017_{10}1710 and 3103_{10}310 to binary:

Dividend (17) = 10001210001_2100012

Divisor (3) = 001120011_200112

- Steps:

1. Align the binary divisor under the dividend.

2. Subtract the divisor from the most significant bits of the dividend. If the result is positive, set a 1 in the quotient and continue. If negative, add the divisor back (restore), set a 0 in the quotient, and shift.

3. Repeat until all bits of the quotient are filled.

Result: The quotient will represent 17/317 / 317/3 and the remainder the leftover.

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3. **Describe Direct Mapping, Associative Mapping, and Set-Associative Mapping**

Direct Mapping: Each memory block maps to a specific cache line. It's a simple and fast method but can cause conflict misses if multiple memory blocks map to the same line.

Associative Mapping: A memory block can go into any cache line. This offers flexibility but requires additional hardware to search the entire cache, increasing cost and complexity.

Set-Associative Mapping: Divides the cache into sets, where each block can be placed in any line within a specific set. It balances speed and flexibility, reducing conflict misses compared to direct mapping.

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4. **Multiply Using Booth's Algorithm (A = 010111, B = 110110)**

Step 1: Convert both numbers to two's complement if needed.

Step 2: Apply Booth's algorithm:

Initialize the registers, set the multiplier (B) and multiplicand (A), and follow Booth's encoding to minimize operations.

Execute add/subtract operations based on the bit pairs, and shift.

Result is obtained after completing the cycles, with the product stored in the accumulator.

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5. **Explain the Non-Restoring Algorithm and Divide (111)2(111)_2(111)2 by (110)2(110)_2(110)2**

Non-restoring division involves conditional subtracting and restoring. For dividing 1112111_21112 by 1102110_21102:

Initialize the divisor and align it with the dividend.

Perform subtraction if the remainder is positive, or add (restore) if it's negative.

Shift left and continue until the full quotient is obtained.

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6. **Multiply -7 and +3 Using Booth's Algorithm with Register Size 4**

Represent -7 and +3 in 4-bit binary:

> -*7* in 4-bit two's complement: 1001
>
> +*3* in 4-bit binary: 0011

Apply Booth's algorithm:

> Initialize the multiplier and multiplicand with these binary values.
>
> For each step, add/subtract the multiplicand from the accumulator based on the Booth's encoding for bit pairs, then shift.
>
> Final result shows the product of -7 and +3 in binary format.